

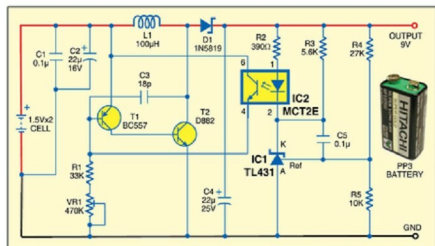
9V CONVERTER USING TWO AA CELLS

■ T.A. BABU

Normally, 9V PP3 batteries last less than a couple of hours if used continuously. These cannot be used if the voltage is below 7.8V. Say goodbye to these expensive

9V batteries having a short life. Two AA-size alkaline cells in conjunction with this DC-DC converter give 9V at a low current and last much longer.

The power supply is designed using a boost converter with fixed 'on' time and variable 'off' time. The



variable 'off' time regulates power to the load. The converter consists of transistor T2, inductor L1 and capacitor C2. The conduction of transistor T1 controls 'off' time of the oscillator in conjunction with capacitor C2. IC TL431 (IC1) monitors the voltage across capacitor C4. When the voltage exceeds 2.5V at the reference pin (Ref) of IC1, the opto-coupler conducts more and reduces the conduction of transistor T1.

The frequency of oscillations mainly depends on the time constant (R-C) of feedback capacitor C3 and the input stage impedance (R1 plus VR1). Adjust preset VR1 to tweak the circuit for efficiency. The converter works with a single cell also. In that case, keep the output current-drain minimal.

EFY note. We have measured maximum output of 8.7V at 28mA current. Above this current, the output becomes zero. ●

DOOR KNOCK ALARM WITH TIMER

■ **PRADEEP G.**

This low-cost circuit uses the piezoelectric element of a piezobuzzer as the input sensor. The piezoelectric element plate is fixed at the centre of the door wing.

by using a cello tape. Apply a small quantity of adhesive at the edges between the plate and the door. Extend wires about 1-1.5 metres from the piezoelectric to the circuit.

IC NE555 (IC1) is configured in monostable mode. When it gets an

input pulse its output goes high for a period set by VR1, resistor R5 and capacitor C3. IC UM66 (IC2) is used as a melody generator. When the door is knocked at, the piezo plate generates an input pulse, which is amplified by transistor T1.

The amplified signal triggers the timer IC NE555 and its output pin 3 goes high to enable the melody generator. Music is heard from the speaker LS1. After the set time period, the melody sound stops.

Assemble the circuit on a general-purpose PCB and enclose in a suitable case. Fix the piezo element at the door and place the speaker in a central room inside the house using long wires. The circuit works off 5-12V DC. The music time can be adjusted through VR1 by changing the R-C time constant of the timer. ●

